

Lab Title: SMTP Email Traffic Forensics

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**Course Name: SBT-DF203: Basic
Networking Skills for Digital Forensics**

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Date of Submission: 15th September 2026

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 4: SMTP Email Traffic Forensics

Three-Week Delivery Block 2/3 | 12-18 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 4
Lab Title	SMTP Email Traffic Forensics
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	smtp.pcap supplied sample capture

1. Source Materials and Slide Alignment

- Email Traffic Forensics: SMTP purpose, ports 25/587/465, plaintext commands, STARTTLS/SMTPS distinctions, Base64 encoding, stream reassembly and email-content analysis.
- The slide practical uses smtp.pcap containing Microsoft Outlook traffic and requires timestamps, client, content, ports, IP addresses and MAC addresses.
- The exercise treats credentials as sensitive evidence and requires protected reporting.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical SMTP packet capture has been supplied for analysis. You must determine when the email exchange occurred, identify the client and server, reconstruct SMTP commands and response codes, decode relevant Base64 authentication fields, recover the message content, and document the network path without exposing sensitive data unnecessarily.

3. Learning Outcomes

- Explain the role of SMTP and distinguish plaintext SMTP, STARTTLS and implicit TLS.
- Identify SMTP commands, server response codes and authentication exchanges.
- Reassemble an SMTP TCP stream and reconstruct message headers and body.
- Decode Base64 values in a controlled offline manner.
- Extract timestamps, client software, IP addresses, ports and MAC addresses.
- Assess evidential limitations when encryption is used or capture is incomplete.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Treat any recovered username, password or message content as confidential assessment evidence. Mask sensitive values in screenshots unless the instructor explicitly requires the full value.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Analyze the capture.
Packet tools	Wireshark and tshark	Filter and reconstruct SMTP streams.
Decoder	Python 3 base64 module	Decode Base64 strings offline.
Hashing	sha256sum	Preserve evidence integrity.
Evidence	smtp.pcap	Historical training traffic.
Text editor	Any local editor	Document recovered message safely.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab4
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y wireshark tshark python3
wget -O evidence/smtp.pcap
'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'
cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
```

Screenshot required: smtp.pcap file details, capinfos summary and matching original/working-copy hashes.

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
(kali@kali)-[~]
$ cd ~/SBT-DF203-Lab4
(kali@kali)-[~/SBT-DF203-Lab4]
$ pwd
/home/kali/SBT-DF203-Lab4
(kali@kali)-[~/SBT-DF203-Lab4]
$ find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./screenshots
./scripts
./reports
(kali@kali)-[~/SBT-DF203-Lab4]
$
```

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab4]
$ wget -O evidence/smtp.pcap 'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'
--2026-09-15 21:16:55-- https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 104.26.10.240, 2606:4700:20::681a:af0, 2606:4700:20::ac43:4b27, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.10.240|:443 ... connected.
HTTP request sent, awaiting response ... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap [following]
--2026-09-15 21:17:10-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap      100%[=====>] 27.20K  --.-KB/s   in 0.009s

2026-09-15 21:17:24 (3.06 MB/s) - 'evidence/smtp.pcap' saved [27850/27850]
```

```

(kali@kali)-[~/SBT-DF203-Lab4]
$ cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap

(kali@kali)-[~/SBT-DF203-Lab4]
$ sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap

(kali@kali)-[~/SBT-DF203-Lab4]
$ capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
File name:          evidence/smtp.pcap
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  60
File size:          27 kB
Data size:          26 kB
Capture duration:   9.198384 seconds
Earliest packet time: 2009-10-05 07:06:07.492060
Latest packet time:  2009-10-05 07:06:16.690444
Data byte rate:     2,920 bytes/s
Data bit rate:      23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256:             17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1:               3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order:  True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60

(kali@kali)-[~/SBT-DF203-Lab4]
$

```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab4-Samuel Chinedu Onu
Trainee name	Samuel Chinedu Onu
Date and time started	15 th September 2026
Evidence file name(s)	smtp.pcap
Source or generation method	https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Original SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
Working-copy SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
Analysis workstation/VM	Kali Linux VM
Notes on any changes	

8. Part A - Inventory the Capture and Locate SMTP Streams

```

tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt

tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
  | tee reports/smtp_packet_inventory.tsv

```

Record the SMTP server port, client ephemeral port, first packet time, last packet time and number of SMTP frames.

```

kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
Warning: program compiled against libxml 215 using older 214

TCP Conversations
Filter:<No Filter>

ive | Duration | | | | Total | Relat
rt | | | Frames Bytes | Frames Bytes | Frames Bytes | Sta
10.10.1.4:1470 ↔ 74.53.140.153:25 25 1,980 bytes 28 22 kB 53 24 kB 0.0
36986000 7.5777

```

```

(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
  | tee reports/smtp_packet_inventory.tsv

Warning: program compiled against libxml 215 using older 214
6      2009-10-05T07:06:08.219663000+0100      74.53.140.153      25      10.10.1.4      1470      S: 220-xc90.websitewelcom
e.com SMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize the use of this system to transport unsol
icited, | and/or bulk e-mail.
7      2009-10-05T07:06:08.224809000+0100      10.10.1.4      1470      74.53.140.153      25      C: EHLO GP
9      2009-10-05T07:06:08.566183000+0100      74.53.140.153      25      10.10.1.4      1470      S: 250-xc90.websitewelcom
e.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10     2009-10-05T07:06:08.568729000+0100      10.10.1.4      1470      74.53.140.153      25      C: AUTH LOGIN
11     2009-10-05T07:06:08.911081000+0100      74.53.140.153      25      10.10.1.4      1470      S: 334 VXNlcm5hbWU6
12     2009-10-05T07:06:08.911655000+0100      10.10.1.4      1470      74.53.140.153      25      C: User: Z3VycGFydGFwQHBh
dHJpb3RzLmLu
13     2009-10-05T07:06:09.253544000+0100      74.53.140.153      25      10.10.1.4      1470      S: 334 UGFzc3dvcmQ6
14     2009-10-05T07:06:09.254118000+0100      10.10.1.4      1470      74.53.140.153      25      C: Pass: cHVuamFiQDEyMw==
15     2009-10-05T07:06:09.613798000+0100      74.53.140.153      25      10.10.1.4      1470      S: 235 Authentication suc
ceeded
16     2009-10-05T07:06:09.614414000+0100      10.10.1.4      1470      74.53.140.153      25      C: MAIL FROM: <gurpartap@
patriots.in>
17     2009-10-05T07:06:09.956765000+0100      74.53.140.153      25      10.10.1.4      1470      S: 250 OK
18     2009-10-05T07:06:09.957250000+0100      10.10.1.4      1470      74.53.140.153      25      C: RCPT TO: <raj_deol2002
in@yahoo.co.in>
19     2009-10-05T07:06:10.319708000+0100      74.53.140.153      25      10.10.1.4      1470      S: 250 Accepted
20     2009-10-05T07:06:10.320203000+0100      10.10.1.4      1470      74.53.140.153      25      C: DATA
21     2009-10-05T07:06:10.661679000+0100      74.53.140.153      25      10.10.1.4      1470      S: 354 Enter message, end
ing with "." on a line by itself
22     2009-10-05T07:06:10.692743000+0100      10.10.1.4      1470      74.53.140.153      25      C: DATA fragment, 1460 by
tes
23     2009-10-05T07:06:10.692786000+0100      10.10.1.4      1470      74.53.140.153      25      C: DATA fragment, 1460 by
tes
24     2009-10-05T07:06:10.692804000+0100      10.10.1.4      1470      74.53.140.153      25      C: DATA fragment, 1460 by
tes
25     2009-10-05T07:06:10.692823000+0100      10.10.1.4      1470      74.53.140.153      25      [TCP Window Full] C: DATA
fragment, 1460 bytes
26     2009-10-05T07:06:10.695115000+0100      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140.153      25      Destinati
on unreachable (Fragmentation needed)

```

9. Part B - Identify Commands and Response Codes

Sequence	Direction	Command/Code	Meaning	Timestamp
1	Server to client	220	Service ready	2009-10-05T07:06:08.219663000+0100
2	Client to server	EHLO/HELO	Client greeting	2009-10-05T07:06:08.224809000+0100
3	Client to server	AUTH	Authentication method	2009-10-05T07:06:08.568729000+0100
4	Client to server	MAIL FROM	Envelope sender	2009-10-05T07:06:09.614414000+0100
5	Client to server	RCPT TO	Envelope recipient	2009-10-05T07:06:09.957250000+0100
6	Client to server	DATA	Message content begins	2009-10-05T07:06:10.320203000+0100

Sequence	Direction	Command/Code	Meaning	Timestamp
7	Client to server	QUIT	Session termination	2009-10-05T07:06:14.763825000+0100

```
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.response.parameter \
| tee reports/smtp_commands_responses.tsv
```

```
kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst \
-e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.response.parameter \
| tee reports/smtp_commands_responses.tsv
Warning: program compiled against libxml 215 using older 214
6 2009-10-05T07:06:08.219663000+0100 74.53.140.153 10.10.1.4 220 xc90.websitewelco
me.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 ,We do not authorize the use of this system to transport unsoli
cited, ,and/or bulk e-mail.
7 2009-10-05T07:06:08.224809000+0100 10.10.1.4 74.53.140.153 EHLO GP 250 xc90.websitewelco
me.com Hello GP [122.162.143.157],SIZE 52428800,PIPELINING,AUTH PLAIN LOGIN,STARTTLS,HELP
10 2009-10-05T07:06:08.568729000+0100 10.10.1.4 74.53.140.153 AUTH LOGIN
11 2009-10-05T07:06:08.911081000+0100 74.53.140.153 10.10.1.4 334 VXNlcm5hbWU6
12 2009-10-05T07:06:08.911655000+0100 10.10.1.4 74.53.140.153
13 2009-10-05T07:06:09.253544000+0100 74.53.140.153 10.10.1.4 334 UGFzc3dvcmQ6
14 2009-10-05T07:06:09.254118000+0100 10.10.1.4 74.53.140.153
15 2009-10-05T07:06:09.613798000+0100 74.53.140.153 10.10.1.4 235 Authentication su
cceded
16 2009-10-05T07:06:09.614414000+0100 10.10.1.4 74.53.140.153 MAIL FROM: <gurpartap@patriots.in>
17 2009-10-05T07:06:09.956765000+0100 74.53.140.153 10.10.1.4 250 OK
18 2009-10-05T07:06:09.957250000+0100 10.10.1.4 74.53.140.153 RCPT TO: <raj_deol2002in@yahoo.co.in>
19 2009-10-05T07:06:10.319708000+0100 74.53.140.153 10.10.1.4 250 Accepted
20 2009-10-05T07:06:10.320203000+0100 10.10.1.4 74.53.140.153 DATA
21 2009-10-05T07:06:10.661679000+0100 74.53.140.153 10.10.1.4 354 Enter message, en
ding with "." on a line by itself
52 2009-10-05T07:06:12.248789000+0100 74.53.140.153 10.10.1.4 250 OK id=1Mugho-0003
Dg-Un
54 2009-10-05T07:06:14.763825000+0100 10.10.1.4 74.53.140.153 QUIT
56 2009-10-05T07:06:15.105467000+0100 74.53.140.153 10.10.1.4 221 xc90.websitewelco
me.com closing connection

(kali@kali)-[~/SBT-DF203-Lab4]
$
```

10. Part C - Decode Base64 Authentication Evidence

Base64 is encoding, not encryption. Copy only the required encoded values into the offline script. Do not use online decoders.

```
python3 - <<'PYLAB'
import base64
samples = {
    'username': 'REPLACE_WITH_CAPTURED_BASE64',
    'password': 'REPLACE_WITH_CAPTURED_BASE64'
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
```

```

kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab4]
$ python3 - <<'PYLAB'
import base64
samples = {
    'username': 'REPLACE_WITH_CAPTURED_BASE64',
    'password': 'REPLACE_WITH_CAPTURED_BASE64'
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
username: decode failed: Invalid base64-encoded string: number of data characters (25) cannot be 1 more than a multiple of 4
password: decode failed: Invalid base64-encoded string: number of data characters (25) cannot be 1 more than a multiple of 4

(kali@kali)-[~/SBT-DF203-Lab4]
$

```

Reporting Rule

In the public-facing report body, show a masked value such as analyst@example.com and P*****3. Place full decoded values only in the protected evidence appendix if required by the instructor.

11. Part D - Reconstruct the Email Message

1. In Wireshark select an SMTP packet and choose Follow > TCP Stream.
2. Identify SMTP envelope fields separately from RFC message headers.
3. Record Date, From, To, Subject, Message-ID, MIME-Version, Content-Type and User-Agent/X-Mailer where present.
4. Determine whether the body is plain text, HTML, multipart or contains an attachment.
5. Save a redacted text reconstruction to reports/reconstructed_email_redacted.txt.

Replace STREAM with the relevant stream number

```
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,STREAM \
| tee reports/smtp_stream_STREAM.txt
```

```
grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt
```

```

(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
| tee reports/smtp_stream_0.txt
Warning: program compiled against libxml 215 using older 214

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHBhdHJpb3RzLmLu

18
334 UGFzc3dvcmQ6

18
cHVuamFiQDEyMw==

```

```

(kali@kali)-[~/SBT-DF203-Lab4]
$ grep -Ei '^([Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer]):' \
reports/smtp_stream_0.txt | tee reports/message_headers.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

(kali@kali)-[~/SBT-DF203-Lab4]
$ cat reports/message_headers.txt
cat reports/smtp_stream_0.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

```

12. Part E - Determine Client, Hosts and Network Metadata

```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv
```

```
tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv
```

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv

Warning: program compiled against libxml 215 using older 214
6      1254722768.219663000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
470    0
7      1254722768.224809000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1470    74.53.140.153    2
5      0
9      1254722768.566183000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
470    0
10     1254722768.568729000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1470    74.53.140.153    2
5      0
11     1254722768.911081000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
470    0
12     1254722768.911655000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1470    74.53.140.153    2
5      0
13     1254722769.253544000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
470    0
14     1254722769.254118000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1470    74.53.140.153    2
5      0
15     1254722769.613798000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
470    0
16     1254722769.614414000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1470    74.53.140.153    2
5      0
17     1254722769.956765000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
470    0
18     1254722769.957250000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1470    74.53.140.153    2
5      0
19     1254722770.319708000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
470    0
20     1254722770.320203000    00:e0:1c:3c:17:c2    00:1f:33:d9:81:60    10.10.1.4    1470    74.53.140.153    2
5      0
21     1254722770.661679000    00:1f:33:d9:81:60    00:e0:1c:3c:17:c2    74.53.140.153    25    10.10.1.4    1
```

```
(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv

Warning: program compiled against libxml 215 using older 214
22     0      Line-based text data (47 lines)
26     0      Line-based text data (12 lines)

(kali@kali)-[~/SBT-DF203-Lab4]
$
```

13. Part F - Encryption and Evidential Limitations

- Check whether STARTTLS is requested and whether subsequent payload becomes TLS.
- If traffic is encrypted, document metadata that remains visible: endpoints, ports, timing, certificate/session information and byte counts.
- Do not claim message content or credentials when the capture does not reveal them.
- Explain that port number alone does not prove encryption; protocol negotiation and packet dissection must be examined.

```

kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst \
-e smtp.req.command -e smtp.req.parameter -e smtp.response.code -e _ws.col.Info \
| grep -i -E 'STARTTLS|TLS|220|EHLO|250'
Warning: program compiled against libxml 215 using older 214
6      2009-10-05T07:06:08.219663000+0100      74.53.140.153      10.10.1.4      220      S: 220-xc90.websi
tewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize the use of this system to transpo
rt unsolicited, | and/or bulk e-mail.
7      2009-10-05T07:06:08.224809000+0100      10.10.1.4      74.53.140.153      EHLO      GP      C: EHLO GP
9      2009-10-05T07:06:08.566183000+0100      74.53.140.153      10.10.1.4      250      S: 250-xc90.websi
tewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
17     2009-10-05T07:06:09.956765000+0100      74.53.140.153      10.10.1.4      250      S: 250 OK
18     2009-10-05T07:06:09.957250000+0100      10.10.1.4      74.53.140.153      RCPT      TO: <raj_deol2002in@yahoo.co.in>C
: RCPT TO: <raj_deol2002in@yahoo.co.in>
19     2009-10-05T07:06:10.319708000+0100      74.53.140.153      10.10.1.4      250      S: 250 Accepted
52     2009-10-05T07:06:12.248789000+0100      74.53.140.153      10.10.1.4      250      S: 250 OK id=1Mug
ho-0003Dg-Un
54     2009-10-05T07:06:14.763825000+0100      10.10.1.4      74.53.140.153      QUIT      C: QUIT

(kali@kali)-[~/SBT-DF203-Lab4]
$

```

During the analysis, I noticed that the server included STARTTLS in its EHLO response, meaning it was providing encryption. But I noticed in the whole communication that the client never actually sent the STARTTLS command. Because of this, the whole session stayed in clear text.

14. Required Findings Worksheet

Question	Finding
When did the SMTP session start and end?	Start: 2009-10-05 07:06:08.219 End: 2009-10-05 07:06:15.105
Client IP/MAC and port	IP: 10.10.1.4 MAC: 00:e0:1c:3c:17:c2 (CradlePoint) Port: 1470
Server IP/MAC and port	IP: 74.53.140.153 MAC: (check eth.dst on an SMTP packet) Port: 25
SMTP server banner	220 xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
Client software	Microsoft Office Outlook 12.0
Authentication method	AUTH LOGIN (Base64 encoded)
Envelope sender and recipient	Sender: gurpartap@patriots.in Recipient: raj_deol2002in@yahoo.co.in
Message From/To/Subject	From: Gurpartap Singh To: raj_deol2002in@yahoo.co.in Subject: SMTP
Message body type	Plain text and HTML
Attachment present?	Yes (NEWS.txt)
STARTTLS/TLS observed?	STARTTLS was provided by the server but not used, the session remained in plaintext
Key limitations	There was no encryption, content of the mail was visible